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CHAPTER 1 • NUMBER • TOPIC 9

Rounding & Bounds

Every rounded number hides a range. Half a unit either side gives the bounds — then pick the bound that makes the answer biggest (or smallest).



$$\frac{3}{10} = 0.3 =$$

30%

6

VARIANTS

26

PAST-PAPER PARTS

50

PRACTICE QUESTIONS

YOU WILL MASTER

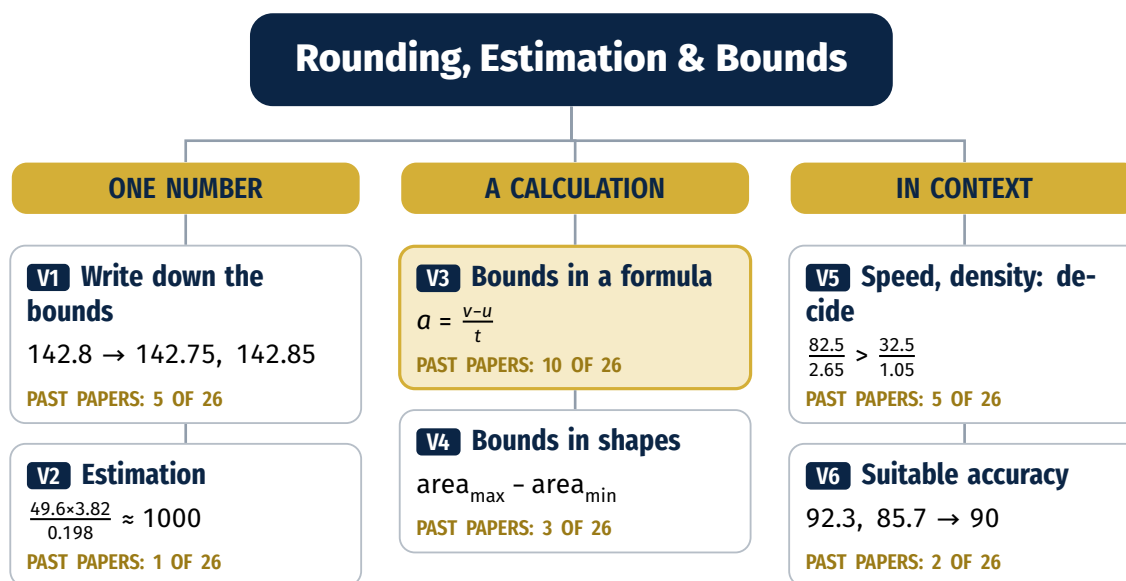
V1 Write down the bounds **V2** Estimation **V3** Bounds in a formula

V4 Bounds in shapes **V5** Speed, density: decide **V6** Suitable accuracy

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Rounding & Bounds at a glance



Your notes
**THE RULE****BIG ÷ SMALL = BIGGEST**

Max of a fraction: biggest top, smallest bottom.

★ How often does each variant come up?

25 past-paper questions (Jan 2020 → Nov 2025) hold 26 parts; 50 practice parts — grouped by type.

	PAST PAPERS (OUT OF 26)	PRACTICE PARTS
V1 Write down the bounds	5	14
V2 Estimation	1	13
V3 Bounds in a formula	10	8
V4 Bounds in shapes	3	3
V5 Speed, density: decide	5	12
V6 Suitable accuracy	2	3

WHAT THIS TELLS YOU A bounds question is on almost every paper: the 3-mark “upper bound of $P = \dots$ ” (V3) is the most common. Learn one table — which bound for each operation — and it is routine.

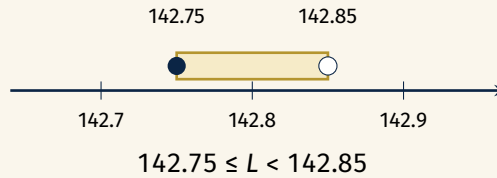
BEFORE YOU START

The big idea

Half a unit either side.

“142.8 m to 1 d.p.” could be anything that rounds to 142.8: from **142.75** up to (but not including) **142.85**.

The unit is 0.1, so go **half of 0.1** = 0.05 each way.

**Which bound do I use?**

$$a + b$$

$$\max = a_U + b_U$$

$$\min = a_L + b_L$$

$$a - b$$

$$\max = a_U - b_L$$

take away the smallest

$$a \times b$$

$$\max = a_U \times b_U$$

$$\min = a_L \times b_L$$

$$a \div b$$

$$\max = \frac{a_U}{b_L}$$

divide by the smallest

**CALCULATOR FIRST, THEN CHECK**

Write the bound you are using under every letter before you calculate, then type the whole formula with brackets in one go. Round only the final answer (to what the question asks).

 Your notes

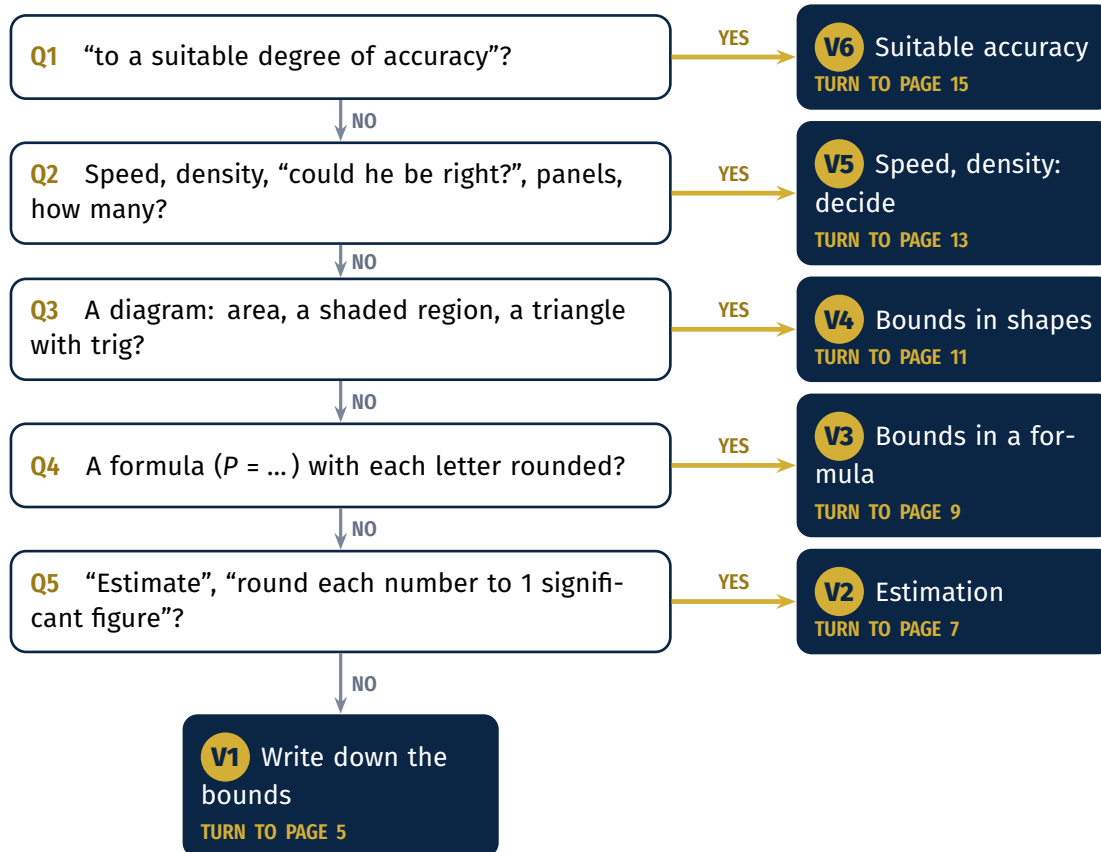
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1

Write Down the Bounds

Half the rounding unit below and above.

SEEN IN PAST PAPERS

5 / 26

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2020 P1HR Q11(a)(b) • Jan 2022 P2H Q4 • Nov 2024 P2H Q2 • Jun 2025 P1HR Q2 • Nov 2025 P1H Q7

? What is it?

In plain words

Find the **unit** it was rounded to (0.1, nearest 0.5, 2 s.f. of 4.3 → 0.1). Take**half** of it. Lower bound = value - half; upper bound = value + half.Error interval: $LB \leq x < UB$.

IT LOOKS LIKE

142.8 (1 d.p.)

18 (nearest 0.5)

4.3 (2 s.f.)



$$18 - 0.25 = 17.75 \quad 18 + 0.25 = 18.25$$

How do I do it?

1 The unit

1 d.p. → 0.1; nearest 5 → 5; 2 s.f. of 4.3 → 0.1.

2 Half it

0.1 → 0.05, 0.5 → 0.25.

3 Bounds

Value ± the half.

4 Interval

 $LB \leq x < UB$ — ≤ on the left, < on the right.

HALF A UNIT EACH WAY

Upper bound: write it as the number, not 142.8499...

V1 Write Down the Bounds

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“correct to 1 decimal place ... write down the lower bound” → **Value** -0.05

“correct to the nearest 0.5 mm” → **Half is 0.25**

“Write down the error interval” → **LB ≤ x < UB**

WORKED EXAMPLE **PAST PAPER**

June 2025 • Paper 1HR • Q2 • 2 marks

The length of a ship is 142.8 m, correct to 1 decimal place. Write down (i) the lower bound (ii) the upper bound.

STEP 1 unit = 0.1 ◀ 1 d.p.

STEP 2 $\frac{1}{2} \times 0.1 = 0.05$ ◀ half it

STEP 3 $142.8 - 0.05 = 142.75$, $142.8 + 0.05 = 142.85$ ◀ bounds

ANSWER (i) 142.75 m (ii) 142.85 m

✓ both round to 142.8 ✓

WATCH OUT

✗ 142.7 and 142.9

✓ 142.75 and 142.85

Half a unit, not a whole unit.

EXAM TIP

For “2 s.f.” find the place value of the 2nd figure: 4.3 → tenths, so ± 0.05 .

Check yourself

calculator first • answers on page 17

1.1 Jan 2022 P2H

A side is 18 mm, correct to the nearest 0.5 mm. Write down the lower and upper bounds.

Answer:

1.2 Jan 2020 P1HR

A cat weighs 4.3 kg to 2 s.f. Write down the upper and lower bounds.

Answer:

1.3 Dr Eslam

$w = 6.4$ kg correct to 1 d.p. Write the error interval for w .

Answer:

V2

VARIANT 2

Estimation

Round every number to 1 significant figure,
then calculate — no calculator.

SEEN IN PAST PAPERS

1 / 26

PRACTICE QUESTIONS: 13


Your notes

Came up in: Nov 2025 P1H Q2

? What is it?**In plain words**

Round each number to

1 significant figure: $49.6 \rightarrow 50$,
 $3.82 \rightarrow 4$, $0.198 \rightarrow 0.2$. Then the sum
is easy in your head.

“Overestimate or underestimate?” —
look at which way you rounded the
numbers.

IT LOOKS LIKE

$$\frac{49.6 \times 3.82}{0.198}$$

$$\sqrt{8.9 \times 4.1}$$

$$\frac{49.6 \times 3.82}{0.198}$$

↓ 1 S.F. EACH

$$\frac{50 \times 4}{0.2} = 1000$$

How do I do it?**1 1 s.f.**

Round every number to one significant figure.

2 Write it

Show the rounded calculation — that is the mark.

3 Work it out $\div 0.2$ is $\times 5$; $\div 0.02$ is $\times 50$.**4 Over or under?**

Numbers on top rounded up $\rightarrow$ over; bottom rounded up
 $\rightarrow$ under.

1 S.F., THEN CALCULATE

Write the rounded numbers.

V2 Estimation

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Work out an estimate for ...” → **1 s.f. each, show it, calculate**

“Is your answer an overestimate?” → **Which way did each number move?**

Division by 0.2 or 0.05 → **Multiply by 5 or by 20**

WORKED EXAMPLE DR ESLAM

Estimation • 3 marks

Work out an estimate for $\frac{49.6 \times 3.82}{0.198}$

STEP 1 $49.6 \rightarrow 50, 3.82 \rightarrow 4, 0.198 \rightarrow 0.2$ ◀ 1 s.f. each

STEP 2 $\frac{50 \times 4}{0.2}$ ◀ write it

STEP 3 $\frac{200}{0.2} = 1000$ ◀ $\div 0.2 = \times 5$

ANSWER about 1000

✓ calculator: 956.9 — close ✓

WATCH OUT

✗ $0.198 \rightarrow 0$

✓ $0.198 \rightarrow 0.2$

1 s.f. keeps the first non-zero digit.

EXAM TIP

The examiner wants the rounded line ($\frac{50 \times 4}{0.2}$) — without it, a correct number scores little.

Check yourself

calculator first • answers on page 17

2.1 Dr Eslam

Estimate $\frac{312 \times 4.1}{59.7}$

Answer:

2.2 Dr Eslam

Estimate $\sqrt{8.9 \times 4.1}$

Answer:

2.3 Dr Eslam

Estimate 0.0496×207

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM

Bounds in a Formula

Choose the bound for each letter that pushes the answer the way you want.

SEEN IN PAST PAPERS

10 / 26

PRACTICE QUESTIONS: 8



Your notes

Came up in: Jan 2020 P1HR Q11(c) · Jan 2020 P1H Q16 · Jan 2021 P1H Q15 · Jan 2021 P1HR Q19 · May 2021 P1H Q20 · Jan 2022 P1H Q18 · Jun 2022 P2H Q18 · Nov 2023 P2H Q17 · Jun 2024 P2H Q19 · Nov 2024 P1H Q21

? What is it?

In plain words

For the **biggest** $\frac{v-u}{t}$: the top as big as possible (v up, u **down**), the bottom as small as possible (t down).

Think about each letter: does making it bigger make the answer bigger? Then use its UB.

$$\text{big} \quad \frac{V_U - U_L}{t_L} \quad \text{small}$$

small

= UPPER BOUND OF a

IT LOOKS LIKE

$$a = \frac{v-u}{t}$$

$$p = \frac{2a-c}{d}$$

$$k = \frac{t}{a-h}$$

How do I do it?

1 All bounds

Write LB and UB of every letter.

2 Choose

For the max: big things you add or divide by... use the table on page 3.

3 Substitute

Put the chosen bounds in, brackets round each.

4 Round

To the accuracy asked (often 2 d.p. or 3 s.f.).

MAX: BIG TOP, SMALL BOTTOM

Minus a small number makes it bigger.

V3 Bounds in a Formula

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“Calculate the upper bound for the value of a ”

→ Each letter's bound chosen for max

“lower bound” of a fraction

→ Small top, big bottom

A minus: $v - u$

→ Max: $v_U - u_L$ WORKED EXAMPLE **PAST PAPER**

January 2021 • Paper 1H • Q15 • 3 marks

$a = \frac{v - u}{t}$. $v = 9.6$, $u = 3.8$ (1 d.p.), $t = 1.84$ (2 d.p.). Calculate the upper bound for a , to 2 d.p.

STEP 1 $v_U = 9.65$, $u_L = 3.75$, $t_L = 1.835$ ◀ bounds needed

STEP 2 max : top big ($v_U - u_L$), bottom small (t_L) ◀ choose

STEP 3 $\frac{9.65 - 3.75}{1.835} = \frac{5.9}{1.835} = 3.215...$ ◀ substitute

ANSWER 3.22

✓ every other choice gives less ✓

 WATCH OUT

✗ $\frac{9.65 - 3.85}{1.845}$ (all upper bounds)

✓ $\frac{9.65 - 3.75}{1.835}$

Upper bounds everywhere do not give the biggest answer when you subtract or divide.

 EXAM TIP

10 of the 26 past-paper parts. Write the three bounds you use in a line before substituting — that line earns M1.

 Check yourself

calculator first • answers on page 17

3.1 Jan 2020 P1H

$P = \frac{2a - c}{d}$; $a = 58.4$ (3 s.f.), $c = 20$, $d = 3.6$ (2 s.f.). Upper bound of P (2 d.p.)?

Answer:

3.2 Jan 2021 P1HR

$k = \frac{t}{a - h}$; $t = 14$, $a = 7.8$, $h = 3.4$ (all 2 s.f.). Lower bound of k ?

Answer:

3.3 May 2021 P1H

$P = \frac{t - w}{y}$; $t = 9.7$, $w = 5.9$ (1 d.p.), $y = 3$ (1 s.f.). Upper bound of P ?

Answer:

V4

VARIANT 4

Bounds in Shapes

Area, perimeter, a shaded region or a trig side: the same choices, with a diagram.

SEEN IN PAST PAPERS

3 / 26

PRACTICE QUESTIONS: 3



Your notes

Came up in: Oct/Nov 2020 P1HR Q19 • Nov 2021 P1H Q13 • Jan 2022 P1HR Q22

? What is it?**In plain words**

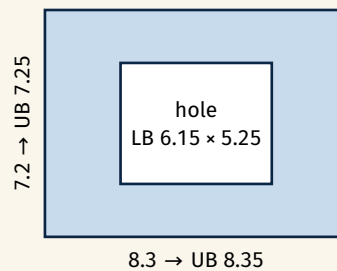
A **shaded region** = big shape – small shape. For its maximum: the **big** one as big as possible, the **small** one as small as possible.

Angles have bounds too: 80° to the nearest degree $\rightarrow 79.5^\circ$ to 80.5° .

IT LOOKS LIKE

rectangle with a hole

sine rule with rounded angles

**How do I do it?****1 Write the formula**

Shaded = big area – small area.

2 Choose bounds

Max: big shape UB, small shape LB.

3 Angles too

Nearest degree $\rightarrow \pm 0.5^\circ$; pick the one that pushes the right way.

4 Calculate

Then round as asked.

MAX REGION = BIG UB – SMALL LB

Draw the diagram with the bounds on it.

V4 Bounds in Shapes

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“upper bound of the area of the shaded region”

→ **Outer UB – inner LB**

Angles “correct to the nearest whole number”

→ $\pm 0.5^\circ$

A triangle side by the sine rule

→ **Choose angle bounds so the fraction is biggest**

WORKED EXAMPLE **PAST PAPER**

Oct/Nov 2020 • Paper 1HR • Q19 • 3 marks

Rectangle $ABCD$ ($AD = 8.3$, $DC = 7.2$) has rectangle $EFGH$ ($EH = 6.2$, $HG = 5.3$) cut out; all 1 d.p. Upper bound of the shaded area?

STEP 1 shaded = $AD \times DC - EH \times HG$ ◀ the formula

STEP 2 8.35, 7.25; 6.15, 5.25 ◀ outer UB, inner LB

STEP 3 $8.35 \times 7.25 - 6.15 \times 5.25 =$
 $60.5375 - 32.2875$ ◀ calculate

ANSWER 28.25 cm^2

✓ biggest outer, smallest hole ✓

WATCH OUT

✗ $8.35 \times 7.25 - 6.25 \times 5.35$

✓ inner LB: 6.15×5.25

Taking away a bigger hole makes the region smaller, not bigger.

EXAM TIP

Label the diagram with the bound you use on each side before calculating.

Check yourself

calculator first • answers on page 17

4.1 Jan 2022 P1HR

Triangle ABC : $c = 11.5$ (1 d.p.), $A = 80^\circ$, $C = 75^\circ$ (nearest degree). Upper bound of b (3 s.f.)?

Answer:

4.2 Dr Eslam

A rectangle is 8 cm by 5 cm (nearest cm). Upper bound of its area?

Answer:

4.3 Dr Eslam

A square has side 6.2 cm (1 d.p.). Lower bound of its perimeter?

Answer:

V5

VARIANT 5

**Speed, Density:
Decide**

Use bounds to test a claim: compare a maximum with a minimum.

SEEN IN PAST PAPERS

5 / 26

PRACTICE QUESTIONS: 12


 Your notes

Came up in: Oct/Nov 2020 P1H Q17 • Jun 2022 P2HR Q18 • May 2023 P1HR Q18 • Jun 2023 P2H Q25 • Jun 2024 P2HR Q19

? What is it?**In plain words**

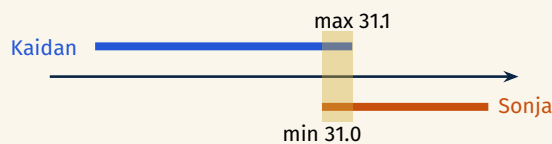
Speed = $\frac{\text{distance}}{\text{time}}$, density = $\frac{\text{mass}}{\text{volume}}$:
 fractions, so max = big top, small
 bottom.

“Could A be more than B?” →
 compare **A's maximum** with
B's minimum.

IT LOOKS LIKE

“could have been greater”

density of a block



The ranges overlap, so Kaidan could be faster.

How do I do it?**1 Bounds**

LB and UB of every measurement.

2 Max of A

Big top, small bottom.

3 Min of B

Small top, big bottom.

4 Decide

A's max > B's min → “could be”. Write both numbers.

MAX OF ONE vs MIN OF THE OTHER Then a sentence.

V5 Speed, Density: Decide

VARIANT 5 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"My average speed could have been greater than ..."

→ **A's max vs B's min**

"upper bound for the density"

→ **Mass UB ÷ volume LB**

"correct to the nearest 5 km"

→ **±2.5**

WORKED EXAMPLE **PAST PAPER**

June 2022 • Paper 2HR • Q18 • 4 marks

Kaidan: 80 km (nearest 5 km) in 2.7 h (1 d.p.). Sonja: 33 km (2 s.f.) in 1 h (nearest 0.1 h). Show that Kaidan's average speed could have been greater than Sonja's.

STEP 1 K: 82.5, 2.65; S: 32.5, 1.05 ◀ bounds needed

STEP 2 K max = $\frac{82.5}{2.65} = 31.13 \dots$ ◀ big top, small bottom

STEP 3 S min = $\frac{32.5}{1.05} = 30.95 \dots$ ◀ small top, big bottom

ANSWER 31.13 > 30.95 — Kaidan could be faster ✓ both numbers shown

WATCH OUT

✗ comparing $\frac{80}{2.7}$ and $\frac{33}{1}$

✓ compare K's max with S's min

Rounded values say "no" — the question is about what is **possible**.

EXAM TIP

Finish with a sentence that quotes both numbers: "31.1 > 30.95, so Kaidan is correct."

Check yourself

calculator first • answers on page 17

5.1 Oct/Nov 2020 P1H

Mass 5 kg (nearest 50 g); volume $1.84 \times 10^{-3} \text{ m}^3$ (3 s.f.). Upper bound of density (1 d.p.)?

Answer:

5.2 Jun 2024 P2HR

Speed 19 km/h (nearest whole), time 1.5 h (1 d.p.). Upper bound of the distance (3 s.f.)?

Answer:

5.3 Dr Eslam

120 km (nearest km) in 1.5 h (1 d.p.). Could the average speed have been over 82 km/h?

Answer:

V6

VARIANT 6

Suitable Degree of Accuracy

Work out both bounds; the answer is the value both round to.

SEEN IN PAST PAPERS

2 / 26

PRACTICE QUESTIONS: 3

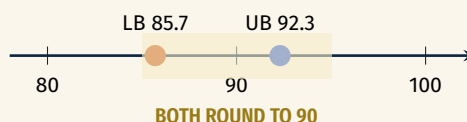
 Your notes

Came up in: Jun 2025 P2H Q22 • Nov 2025 P2H Q24

? What is it?

In plain words

Find the **upper** and **lower** bound of the answer. Round both to 3 s.f., 2 s.f., 1 s.f. ... stop at the most accurate level where they **agree**. That is the answer you can trust.



IT LOOKS LIKE

92.3 and 85.7 → 90

How do I do it?

1 UB of the answer

Choose bounds for the maximum.

2 LB of the answer

Choose bounds for the minimum.

3 Round both

Try 3 s.f., 2 s.f., 1 s.f.

4 Answer

The first level where they match — say why.

BOTH BOUNDS AGREE Explain: "both round to 90".

V6 Suitable Degree of Accuracy

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“By considering bounds, work out ... to a suitable degree of accuracy”

→ **UB, LB, round both until they agree**

Area of a shaded region with rounded sides

→ **Max: big UB, hole LB; min the reverse**

“You must show your working”

→ **Write both bounds to 4+ figures**

**WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 2H · Q22 · 4 marks

Shaded area $X = AB \times BC - (\text{square side})^2$. $AB = 11.5$ (nearest 0.5), $BC = 9.2$ (2 s.f.), square side 4.1 (2 s.f.). Find X to a suitable degree of accuracy.

STEP 1

$$UB = 11.75 \times 9.25 - 4.05^2 = 92.285$$

◀ max: big rectangle, small square

STEP 2

$$LB = 11.25 \times 9.15 - 4.15^2 = 85.715$$

◀ min: the reverse

STEP 3

2 s.f. : 92, 86 (differ); 1 s.f. : 90, 90

◀ round both

ANSWER $X = 90$ (1 s.f.) — both bounds round to 90 ✓ reason given

**WATCH OUT**

× giving 92.3 or 89

✓ the value both bounds round to

A suitable accuracy must hold for every possible value.

**EXAM TIP**

Write the two bounds **and** the sentence “both round to 90 to 1 s.f.” — the last mark needs the reason.

**Check yourself**

calculator first · answers on page 17

6.1

Dr Eslam

$a = 7.4$, $b = 2.3$ (both 1 d.p.). Give $a \times b$ to a suitable degree of accuracy.

Answer:

6.2

Dr Eslam

52 km (nearest km) in 1.3 h (1 d.p.). Speed to a suitable degree of accuracy?

Answer:

6.3

Dr Eslam

A rectangle is 12.4 cm by 8.6 cm (1 d.p.). Area to a suitable degree of accuracy?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 17.75 mm, 18.25 mm

1.2 UB 4.35 kg, LB 4.25 kg

1.3 $6.35 \leq w < 6.45$

2.1 $\frac{300 \times 4}{60} = 20$

2.2 $\sqrt{9 \times 4} = 6$

2.3 $0.05 \times 200 = 10$

3.1 27.44

3.2 3

3.3 1.56

4.1 5.25 cm

4.2 46.75 cm^2

4.3 24.6 cm

5.1 2738.4 kg/m^3

5.2 30.2 km

5.3 yes: UB = $83.1 > 82$

6.1 20 (1 s.f.)

6.2 40 km/h (1 s.f.)

6.3 110 cm^2 (2 s.f.)

THE WHOLE TOPIC ON ONE PAGE

Rounding & Bounds cheat sheet



Your notes
V1 Bounds 5/26

$$142.75 \leq L < 142.85$$

Half a unit each way.

V2 Estimation 1/26

$$\frac{50 \times 4}{0.2} = 1000$$

1 s.f. each; show it.

V3 Formula 10/26

$$\frac{9.65 - 3.75}{1.835}$$

Max: big top, small bottom.

V4 Shapes 3/26

$$8.35 \times 7.25 - 6.15 \times 5.25$$

Outer UB, hole LB.

V5 Decide 5/26

$$31.13 > 30.95$$

Max of one vs min of the other.

V6 Suitable accuracy 2/26

$$92.3, 85.7 \rightarrow 90$$

Round both until they agree.

**TOP 3 LOST MARKS**

1. a whole unit instead of half ×
2. all upper bounds for a max ×
3. no reason for the accuracy ×

BOTTOM LINE Half a unit each way · +, ×: same bound · -, ÷: the opposite bound · decide with max vs min.